Benchmarking Estimators for the Local Average Treatment Effect

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April 13, 2026

Overview

- **Problem.** Estimating causal effects when treatment is endogenous and a valid instrument is available.
- **Estimand.** Local Average Treatment Effect (LATE) — the average effect for compliers.
- **Question.** Which modern causal ML methods outperform classical 2SLS: when and by how much?
- **Methods compared.**
  - Two-Stage Least Squares (2SLS)
  - Post-Lasso IV
  - Double Machine Learning IV (DML-IV)
  - Augmented IPW IV (AIPW-IV)
  - IV Targeted Maximum Likelihood (IV-TMLE)
- **Evaluation.** Bias, RMSE, coverage, and CI width across simulated DGPs varying instrument strength, sample size, and nuisance complexity.

Setup: Potential Outcomes & IV Framework

For each unit $i = 1, \ldots, n$, we observe (Y_i, D_i, Z_i, X_i): outcome, binary treatment, binary instrument, covariates.
Each unit has potential outcomes $Y_i(d)$ and potential treatments $D_i(z)$ for $d, z \in \{0, 1\}$.
Under monotonicity, every unit is one of three types:

- **Always-takers:** $D_i(0) = D_i(1) = 1$
- **Never-takers:** $D_i(0) = D_i(1) = 0$
- **Compliers:** $D_i(0) = 0$, $D_i(1) = 1$

The LATE is identified as the Wald ratio:

$$\tau_{\text{LATE}} = \mathbb{E}[Y_i(1) - Y_i(0) \mid \text{complier}] = \frac{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]}{\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0]}$$

Identifying Assumptions

All five methods rely on the following assumptions *(Imbens & Angrist 1994)*:

1. **SUTVA.** No interference between units.
2. **Independence.** $\{Y_i(d), D_i(z)\} \perp\!\!\!\perp Z_i \mid X_i$ — instrument is as good as randomly assigned given covariates.
3. **Exclusion.** Z_i affects Y_i only through D_i.
4. **Monotonicity.** $D_i(1) \geq D_i(0)$ almost surely — no defiers.
5. **Relevance.** $\mathbb{E}[D_i(1) - D_i(0) \mid X_i = x] \geq \delta > 0$ for all x.
6. **Overlap.** $\eta \leq P(Z_i = 1 \mid X_i = x) \leq 1 - \eta$ for all x.

All five methods target τ_{LATE}: no method requires a stronger structural assumption than the above.

Method 1: Two-Stage Least Squares (2SLS)

The structural model imposes linearity in both stages:

$$Y_i = \tau D_i + \beta^\top X_i + \varepsilon_i, \quad D_i = \pi Z_i + \gamma^\top X_i + \eta_i$$

Stage 1: regress D_i on (Z_i, X_i), obtain $\hat{D}_i$.
Stage 2: regress Y_i on $(\hat{D}_i, X_i)$; coefficient on $\hat{D}_i$ is $\hat{\tau}_{2SLS}$.

Under correct specification, $\sqrt{n}(\hat{\tau}_{2SLS} - \tau_{\mathrm{LATE}}) \xrightarrow{d} \mathcal{N}(0, \Sigma_{2SLS})$.

Strengths: Interpretable, computationally straightforward, efficient under correct specification.

Weaknesses: Linearity of both equations is imposed; any nonlinearity in $g(X_i)$ produces bias. Many-variable bias when p is large or when $p/n \not\to 0$.

Method 2: Post-Lasso IV

For high-dimensional X_i (p comparable to n), 2SLS is infeasible. Post-double-selection Lasso *(Belloni, Chernozhukov & Hansen 2014)* uses the Lasso as a model-selection device.

Three steps:

1. Lasso Y_i on $X_i \to$ selected index set $\hat{I}_Y$.
2. Lasso D_i on $X_i \to$ selected index set $\hat{I}_D$.
3. Run 2SLS using Z_i as instrument and $X_{\hat{I}_Y \cup \hat{I}_D}$ as controls.

The union selection is critical: single selection leads to omitted variable bias.

Strengths: Valid $\sqrt{n}$-inference in the scalar LATE model under approximate sparsity; implemented in `hdm`.

Weaknesses: Inconsistent when $g(X_i)$ is dense; sensitive to penalty λ.

Method 3: Double Machine Learning IV (DML-IV)

Chernozhukov et al. (2018). The partially linear IV model:

$$Y_i = \tau D_i + g(X_i) + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i \mid Z_i, X_i] = 0$$

Regress Y, D, Z on X non-parametrically via cross fitting and obtain residuals $\tilde{Y}_i, \tilde{D}_i, \tilde{Z}_i$. The **Neyman-orthogonal** moment condition:

$$\frac{1}{n}\sum_i \tilde{Z}_i(\tilde{Y}_i - \tau\tilde{D}_i) = 0 \quad \Rightarrow \quad \hat{\tau}_{DML} = \frac{\sum_i \tilde{Z}_i\tilde{Y}_i}{\sum_i \tilde{Z}_i\tilde{D}_i}.$$

Strengths: $\sqrt{n}$-valid inference with any nuisance learner achieving $o(n^{-1/4})$ rate; no sparsity or linearity required; `DoubleML` package.

Weaknesses: Not doubly robust; requires all three nuisance functions to be consistently estimated.

Method 4: AIPW-IV

Tan (2006); Okui et al. (2012). Uses the efficient influence function for the LATE:

$$\phi_i^Y = \frac{(Z_i - e_i)(Y_i - \mu_Y(X_i, Z_i))}{e_i(1 - e_i)} + \mu_Y(X_i, 1) - \mu_Y(X_i, 0)$$

$$\phi_i^D = \frac{(Z_i - e_i)(D_i - \mu_D(X_i, Z_i))}{e_i(1 - e_i)} + \mu_D(X_i, 1) - \mu_D(X_i, 0)$$

$$\hat{\tau}_{AIPW} = \frac{n^{-1} \sum_i \hat{\phi}_i^Y}{n^{-1} \sum_i \hat{\phi}_i^D}$$

Strengths: Doubly robust (consistent if either outcome models or propensity is correctly specified); achieves the semiparametric efficiency bound; valid $\sqrt{n}$-inference.

Weaknesses: Requires estimating three nuisance functions; extreme pseudo-outcomes when $e(X_i) \approx 0$ or 1, or when local compliance is near zero.

Method 5: IV-TMLE

van der Laan & Rose (2011); Zheng & van der Laan (2017).
Same nuisance functions as AIPW-IV. Adds a **targeting step**: fit the clever covariate $H_i = (Z_i - e_i)/[e_i(1 - e_i)\hat{c}(X_i)]$ and update the outcome model:

$$\hat{\mu}_Y^*(x, z) \;=\; \hat{\mu}_Y(x, z) + \hat{\varepsilon}\, H(x, z)$$

where $\hat{\varepsilon}$ solves the efficient score equation. The substitution estimator:

$$\hat{\tau}_{TMLE} = \frac{\overline{\hat{\mu}_Y^*(X, 1) - \hat{\mu}_Y^*(X, 0)}}{\bar{\hat{c}}(X)}$$

Strengths: Same double robustness and efficiency as AIPW-IV; the substitution form **respects parameter bounds** (natural choice for binary/bounded outcomes).

Weaknesses: Requires compliance $\hat{c}(X_i)$ in the clever covariate denominator; Need to estimate three nuisance functions.

Methods at a Glance

Method	Nonpar. nuisance	High-dim	DR	Efficient
2SLS	$\times$	$\times$	$\times$	Under lin.
Post-Lasso	$\times$	$\checkmark$	$\times$	Under sparsity
DML-IV	$\checkmark$	$\checkmark$	$\times$	$\times$
AIPW-IV	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
IV-TMLE	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

The hierarchy: 2SLS (linearity) $\rightarrow$ Post-Lasso (sparsity) $\rightarrow$ DML-IV (nonparametric) $\rightarrow$ AIPW-IV / IV-TMLE (doubly robust + efficient).

Simulation: Data-Generating Process

$n \in \{500, 1000, 2000\}$, $p = 5$ covariates $X_i \sim \mathcal{N}(0, I_5)$.

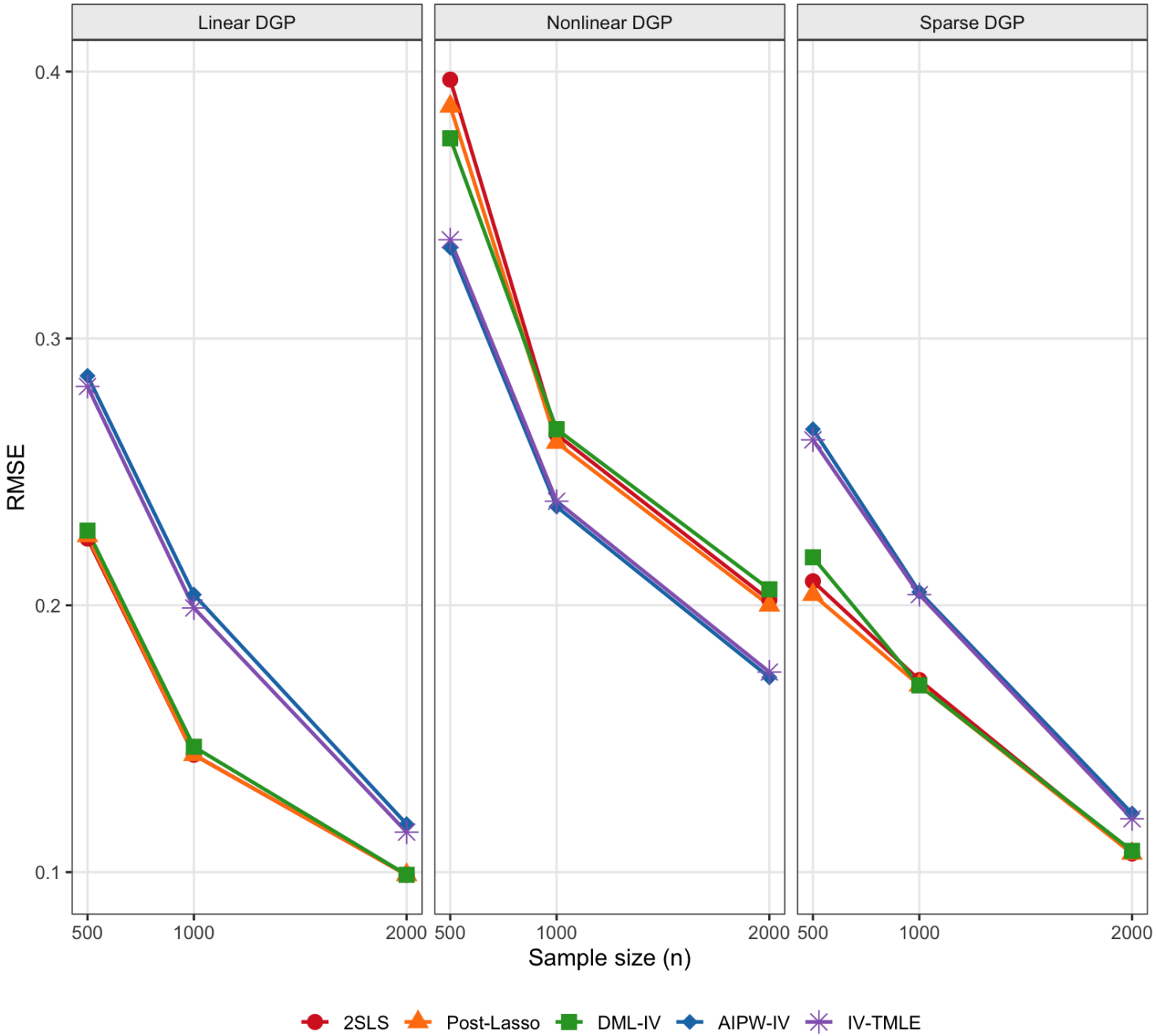
Instrument: $e(X_i) = \sigma(X_{i1} + 0.5X_{i2})$, $Z_i \sim \mathrm{Bern}(e(X_i))$.
Treatment: $D_i(z) \sim \mathrm{Bern}(\sigma(-1 + 0.5X_{i1} + \gamma z))$, $D_i = D_i(Z_i)$.
Instrument strength $\gamma \in \{0.5, 1.0, 2.0\}$; compliance rate increases with IV strength.
Outcome: $Y_i = D_i + g(X_i) + \varepsilon_i$, true LATE $= 1$.

Three nuisance specifications for g:

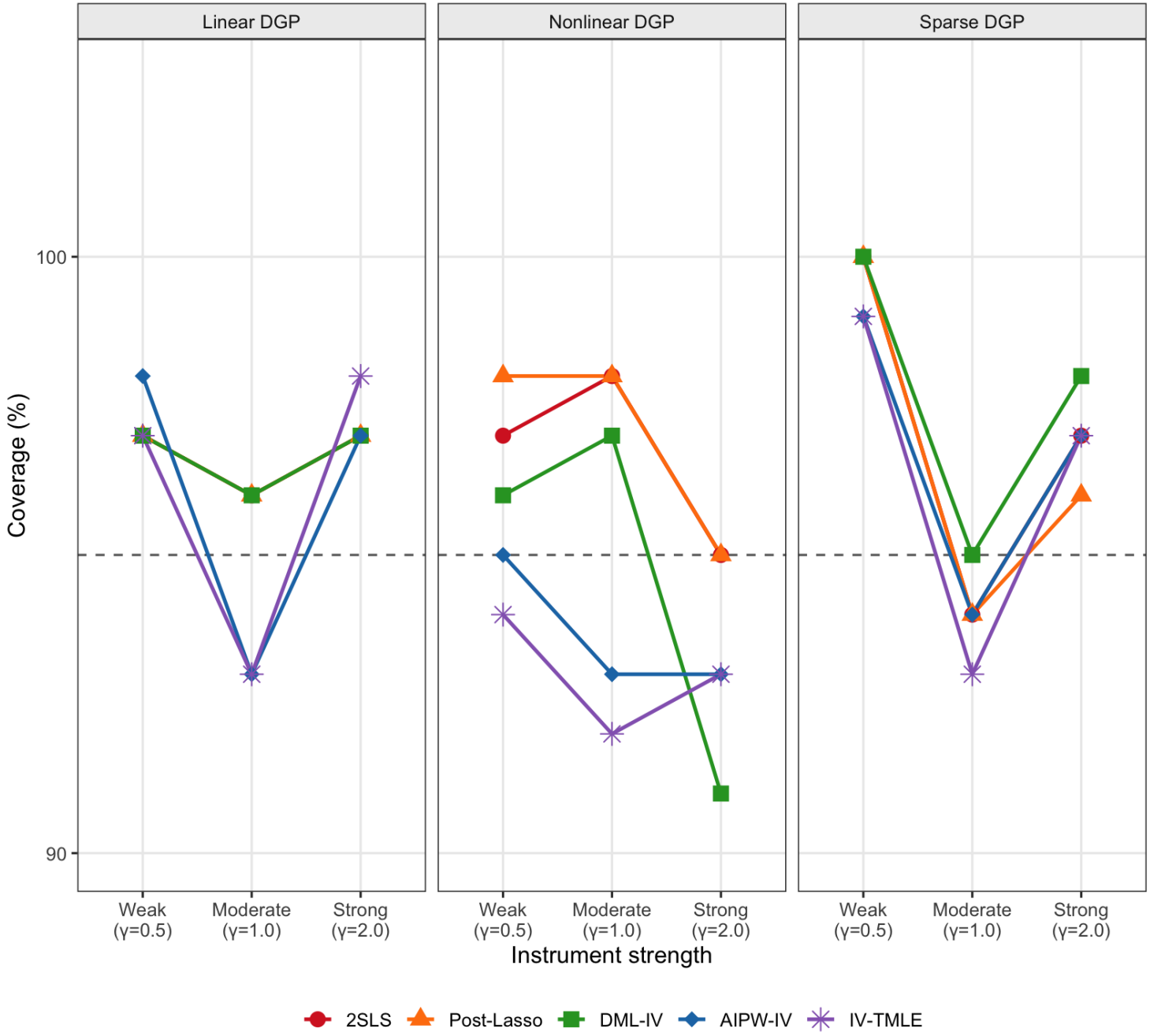
- **Linear:** $X_1 + 0.6X_2 - 0.4X_3 + 0.3X_4 - 0.2X_5$
- **Nonlinear:** $0.5X_1^2 + \sin(\pi X_2) + X_1X_2 - 0.4X_3^2$
- **Sparse:** X_1

$R = 100$ replications per cell; nuisance estimated via cross-fitted Lasso (g, ℓ, e) and random forests (μ_Y, μ_D).

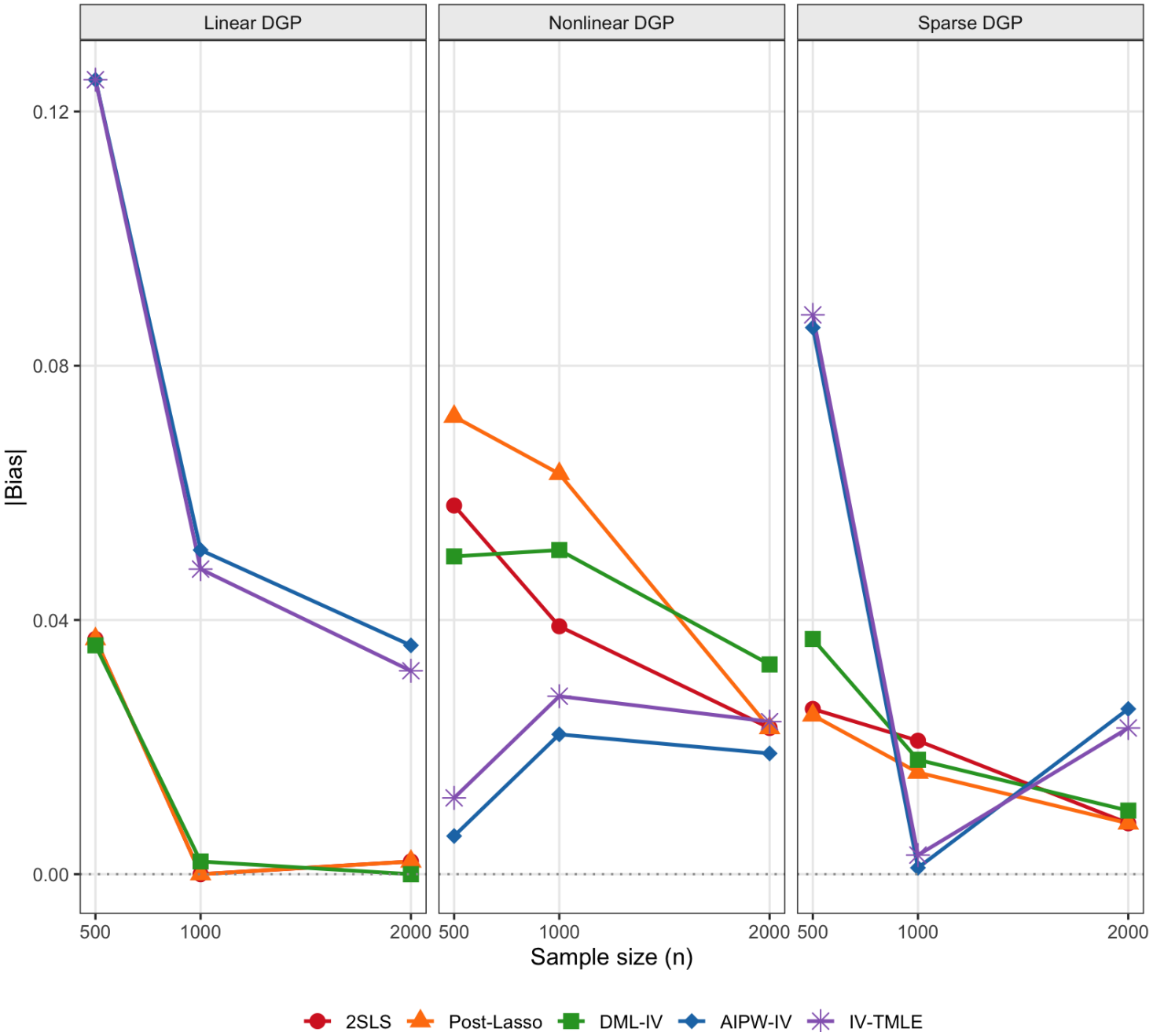
RMSE by Sample Size and DGP ($\gamma = 2, R = 100$)



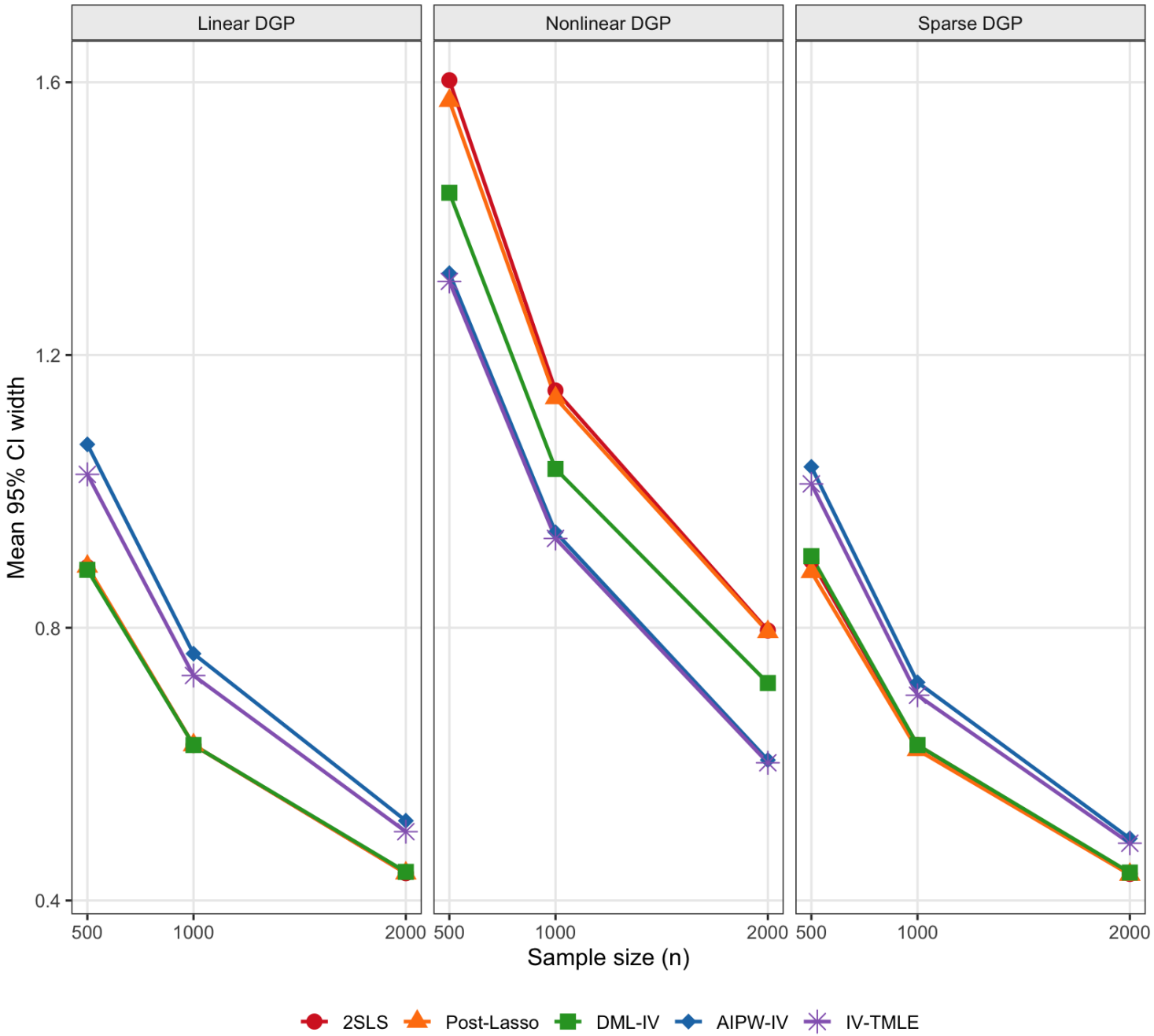
Coverage by Instrument Strength ($n = 2000$)



Bias by Sample Size and DGP ($\gamma = 2$)



CI Width by Sample Size and DGP



Practical Recommendations

- **Always report 2SLS.** Agreement with DML-IV signals linearity is credible; disagreement signals it is not.
- **Use Post-Lasso IV when p is large and sparsity is plausible.** Always apply the double-selection procedure. Never use single-selection Lasso.
- **Use DML-IV as the default causal ML estimator.** No sparsity or linearity required; accommodates any nuisance learner achieving $o(n^{-1/4})$; `DoubleML` package.
- **Use AIPW-IV or IV-TMLE for double robustness.** Prefer IV-TMLE for binary or bounded outcomes. Trim observations with extreme propensity scores.

Limitations & Future Work

Limitations. Single binary instrument, binary treatment only, moderate covariate dimension ($p = 20$), $R = 100$ replications (more needed).

Future work.

- Extension to CLATE estimators (interacted 2SLS, Post-Lasso Interacted IV, IV Forest, IV-DR Learner).
- Right censored survival outcomes.
- Real data application (lottery instrument in Oregon Health Insurance Experiment).

Github repository.

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